C Programming Assignment – Question 1

1. Cab Fare Calculation

A cab service calculates the final fare based on the distance travelled and the customer's membership. The base fare is ■100. If the distance travelled is more than 10 km, an additional ■20 is charged for every kilometer beyond 10 km. Customers with a membership receive a 10% discount on the final fare. Develop a C program to calculate the amount payable.

Sample Input

```
Distance: 15
Member: true
```

Sample Output

```
Final Fare: ■180
```

C Program

```
#include <stdio.h>

int main(void) {
    float distance, fare = 100.0f;
    int member;

    printf("Enter distance travelled (km): ");
    scanf("%f", &distance);

    printf("Is the customer a member? (1 = Yes, 0 = No): ");
    scanf("%d", &member);

    if (distance > 10.0f) {
        fare += (distance - 10.0f) * 20.0f;
    }

    if (member == 1) {
        fare *= 0.90f;  // Apply 10% discount
    }

    printf("Final Fare: ■%.2f\n", fare);
    return 0;
}
```

Explanation

1. Start with a base fare of ■100.
2. If the distance is greater than 10 km, add ■20 for each kilometer beyond 10 km.
3. If the customer is a member, reduce the fare by 10%.
4. Print the final fare.

Note: The sample output in the assignment says ■180, but applying the stated 10% membership discount to a 15 km trip gives a fare of ■180 before discount and ■162 after discount. The program follows the problem statement.